

EMU Servo Tuner

Bench Tool User Guide

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KJO Liquid Rocket Motor Control System

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A Serial-console tool, flashed onto the actual EMU hardware, for interactively tuning the PWM open/close and potentiometer open/close limits of each of EMU's ball valves.

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1 Purpose

EMU Servo Tuner is a standalone sketch, built and flashed onto the same physical EMU hardware (Feather M0 + Servo FeatherWing) as production firmware, that lets an operator select one of a small built-in list of valves and interactively tune its PWM/encoder limits over Serial – the same target/adjust workflow as the standalone **Servo Tuner**, but driven by a per-valve table instead of one hardcoded channel/pin pair.

Warning

The valve table built into this tool (§3) is a snapshot taken when the tool was written – it is **not** automatically kept in sync with EMU’s production `KJO_Valve.h`. Treat the printed limits as a starting point only, and always verify against the current production reference doc (*KJO EMU Reference*) before trusting a number from this tool’s menu. In particular, this tool’s built-in list still includes a **FILL** entry; production EMU has no Fill valve of its own (it lives on GSEMU) – the entry’s numbers actually match GSEMU’s Fill valve calibration, not an EMU valve.

2 Hardware Required

- EMU hardware: Adafruit Feather M0 + PWM/Servo FeatherWing (PCA9685)
- The valve(s) to be tuned, wired to their normal EMU pin/channel assignments
- USB cable for Serial (115200 baud)

3 Built-In Valve Table

Valve	Servo ch.	PWM open	PWM close	Pot pin	Encoder open/closed
FILL	4	2329	3918	A4	1310 / 3004
OX FEED	5	2050	3865	A3	1146 / 2846
FUEL FEED	6	2280	3950	A1	1217 / 2887

These are the limits displayed when a valve is selected – they are the tool’s own starting reference, not necessarily what is currently flashed to production EMU.

4 How to Use

Open a Serial monitor at 115200 baud after powering the board.

4.1 Outer Loop – Valve Selection

1. The tool prints a numbered menu of the valves in §3. Enter the number and press Enter.
2. It prints that valve’s servo channel, pot pin, and current PWM/encoder open-close limits.

4.2 Inner Loop – Target / Adjust

Identical workflow to the standalone Servo Tuner, using the selected valve’s own channel and pot pin:

1. **Target:** enter a PWM count (the valve’s own open/close values are printed as a range hint).

2. **Hysteresis delta:** enter a count value; the tool approaches the target from **target + delta** counts away to remove backlash from the reading.
3. The servo moves and the resulting potentiometer reading is printed.
4. **Action prompt** – type one letter and press Enter:
 - **a** (adjust): enter a signed delta; the tool moves away from the *previous* target by the hysteresis amount, then to the new target, and prints the new pot reading.
 - **c** (continue): return to step 1 for a fresh target on the *same* valve.
 - **f** (finished): return to the outer valve-selection menu.

5 Notes

- The Serial console has no timeout – the tool waits indefinitely at every prompt.
- Backspace is supported while typing a value.
- An unrecognised action-prompt letter simply re-prints the prompt; it does not exit or lose the current target.

6 Version History

Version	Date	Notes
0.1	September 2026	Initial user guide.
